



Features

- 150 mA current capability
- Very high tracking accuracy
- Output voltage adjustable down to 2.0 V
- Stable with ceramic output capacitors
- Very low dropout voltage of typically 250 mV at 150 mA
- Very low current consumption of typically 0.1 μ A in standby mode
- Power Good output indicates overvoltage and undervoltage
- Internally controlled soft start
- Wide input voltage range: $-16\text{ V} \leq V_{\text{IN}} \leq 45\text{ V}$
- Wide temperature range: $-40\text{ }^{\circ}\text{C} \leq T_{\text{J}} \leq 150\text{ }^{\circ}\text{C}$
- Short circuit protected output (to GND and to battery)
- Reverse polarity protected input
- Overtemperature protection
- Green Product (RoHS compliant)
- AEC-Q100 qualified

Description

The HE115(J) is a monolithic integrated low-dropout voltage tracking regulator with high accuracy in small ESOP-8 packages and DFN-10L packages.

The HE115(J) is designed to supply off-board systems, for example sensors in powertrain management systems under the severe conditions of automotive applications.

Therefore, the HE115(J) is equipped with protection functions against reverse polarity and against short circuit to GND and battery up to a supply voltage of 45 V and output currents up to 150 mA. The output voltage follows the reference voltage that is applied to the ADJ input with very high accuracy. The required minimum reference voltage at ADJ is 2.0 V.

Applications

- Automotive sensor supply
- Protected sensor supply for off-board sensors
- Secondary voltage supply in automotive ECU
- High precision voltage tracking
- Precision voltage replication
- Power switch for off-board load

Part Number	Package	Marking	Packing
HE115RBQA	ESOP-8	HE115	4000 / Tape & Reel
HE115JRJAA	DFN-10L	HE115J	4000 / Tape & Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

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1 Block diagram and pin description

1.1 Block diagram

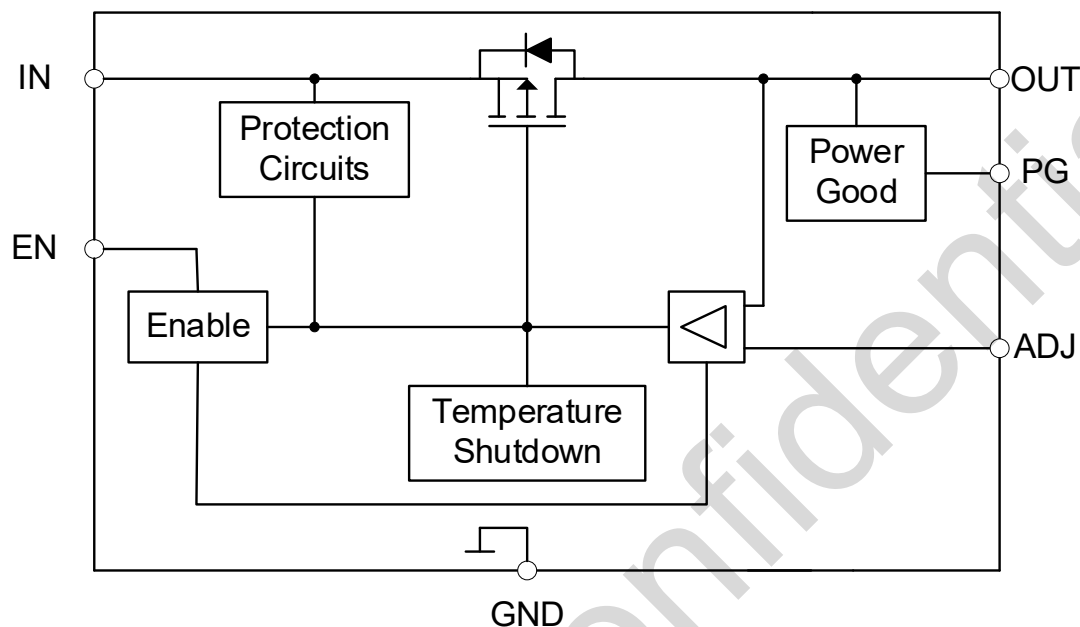


Figure 1-1 Block diagram

2 Pin configuration

2.1 Pin assignment(ESOP-8)

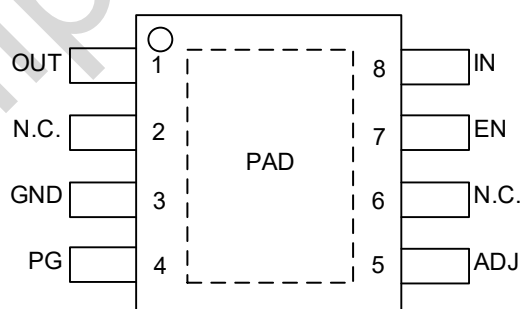


Figure 2-1 ESOP-8 Package

Table 2-1 Pin description(ESOP-8)

Pin #	Symbol	Function	Description
1	OUT	Tracker output	Tracker output. 150 mA output current capability. Connect to GND with a capacitor close to the pins, respecting capacitance and ESR requirements given in Table 3-2 Functional range.
2,6	NC		Not connected.
3	GND		Ground.
4	PG	Power Good	Open Drain output. Connect via a pull-up resistor to a positive voltage rail. A “low” signal indicates a fault condition of the tracker output.
5	ADJ	Adjust	Connect to the reference voltage
7	EN	Enable input	“High” signal : enables the tracker. “Low” signal : disables the tracker. If the enable function is not required, then connect EN to IN.
8	IN	Input	It is recommended to connect a small ceramic capacitor to GND close to the pins in order to compensate line influence.
	Pad		Exposed Pad.

2.2 Pin assignment(DFN-10L)

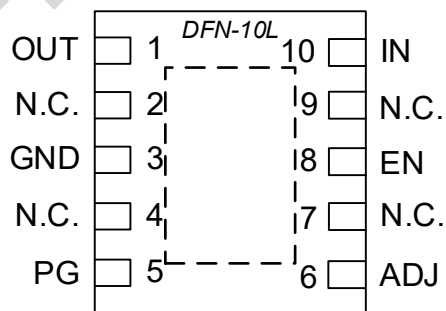


Figure 2-2 DFN-10L Package

Table 2-2 Pin description(DFN-10L)

Pin #	Symbol	Function	Description
1	OUT	Tracker output	Tracker output. 150 mA output current capability. Connect to GND with a capacitor close to the pins, respecting capacitance and ESR requirements given in Table 3-2 Functional range.
2,4,7,9	NC		Not connected.
3	GND		Ground.
5	PG	Power Good	Open Drain output. Connect via a pull-up resistor to a positive voltage rail. A “low” signal indicates a fault condition of the tracker output.
6	ADJ	Adjust	Connect to the reference voltage
8	EN	Enable input	“High” signal : enables the tracker. “Low” signal : disables the tracker. If the enable function is not required, then connect EN to IN.
10	IN	Input	It is recommended to connect a small ceramic capacitor to GND close to the pins in order to compensate line influence.
	Pad		Exposed Pad.

3 Electrical specifications

3.1 Absolute maximum ratings

$T_j = -40\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Table 3-1 Absolute maximum ratings⁽¹⁾

Symbol	Parameter	Value		Unit
		Min	Max	
V_{IN}	Input voltage	-16	45	V
V_{EN}	Enable voltage	-16	45	V
V_{ADJ}	Adjust voltage	-16	45	V
V_{OUT}	Output voltage	-5	45	V
V_{PG}	Power Good voltage	-0.3	20	V
T_j	Junction temperature	-40	150	$^{\circ}\text{C}$
T_{STG}	Storage temperature	-55	150	$^{\circ}\text{C}$
$V_{ESD,HBM}$	ESD susceptibility to GND (Human Body Model (HBM)) ⁽²⁾	-4000	4000	V
$V_{ESD,CDM}$	ESD susceptibility to GND (Charged Device Model (CDM)) ⁽³⁾	-750	750	V

Note:

Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods of time may affect device reliability.

Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as outside the normal operating range. Protection functions are not designed for continuous repetitive operation.

(1). Not subject to production test, specified by design.

(2). ESD susceptibility, Human Body Model (HBM) according to ANSI/ESDA/JEDEC JS-001 (1.5 k Ω , 100 pF).

(3). ESD susceptibility, Charged Device Model (CDM) according to JEDEC JESD22-C101.

3.2 Functional range

Table 3-2 Functional range

Symbol	Parameter	Value			Unit
		Min	Typ	Max	
V _{IN}	Input voltage range	4	-	45	V
V _{ADJ}	Adjust input voltage range (voltage tracking range)	2	-	14	V
C _{OUT}	Capacitance of output capacitor ⁽¹⁾⁽²⁾	1	-	-	uF
ESR _{COUT}	Output capacitor's Equivalent Series Resistance ⁽²⁾	-	-	5	Ω
T _J	Junction temperature ⁽¹⁾	-40	-	150	°C

Note:

Within the functional range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the electrical characteristics table.

(1). The minimum output capacitance requirement is applicable for a worst case capacitance tolerance of 30%.

(2). Not subject to production test, specified by design.

3.3 Thermal data

Table 3-3 Thermal data in ESOP-8

Symbol	Parameter	Value			Unit
		Min	Typ	Max	
R _{thJC}	Junction to case ⁽¹⁾	-	17	-	K/W
R _{thJP}	Junction to pin ⁽¹⁾	-	96	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (2s2p board ⁽²⁾)	-	67	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, footprint only ⁽³⁾)	-	194	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, 300mm ² heatsink area on PCB ⁽³⁾)	-	82	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, 600mm ² heatsink area on PCB ⁽³⁾)	-	68	-	K/W

(1). Not subject to production test, specified by design.

(2). Specified R_{thJA} value is according to JEDEC JESD51-2,-5,-7 at natural convection on FR4 2s2p board; the product (chip and package) was simulated on a 76.2 × 114.3 × 1.5 mm³ board with two inner copper layers (2 × 70 μm Cu, 2 × 35 μm Cu). Where applicable, a thermal via array next to the package contacted the first inner copper layer.

(3). Specified R_{thJA} value is according to JEDEC JESD51-3 at natural convection on FR4 1s0p board; the product (chip and package) was simulated on a 76.2 × 114.3 × 1.5 mm³ board with one copper layer (1 × 70 μm Cu).

Table 3-4 Thermal data in DFN-10L

Symbol	Parameter	Value			Unit
		Min	Typ	Max	
R _{thJC}	Junction to case ⁽¹⁾	-	18	-	K/W
R _{thJP}	Junction to pin ⁽¹⁾	-	85	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (2s2p board ⁽²⁾)	-	50	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, footprint only ⁽³⁾)	-	157	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, 300mm ² heatsink area on PCB ⁽³⁾)	-	77	-	K/W
R _{thJA}	Junction to ambient ⁽¹⁾ (1s0p board, 600mm ² heatsink area on PCB ⁽³⁾)	-	63	-	K/W

(1). Not subject to production test, specified by design.

(2). Specified R_{thJA} value is according to JEDEC JESD51-2,-5,-7 at natural convection on FR4 2s2p board; the product (chip and package) was simulated on a 76.2 × 114.3 × 1.5 mm³ board with two inner copper layers (2 × 70 μm Cu, 2 × 35 μm Cu). Where applicable, a thermal via array next to the package contacted the first inner copper layer.

(3). Specified R_{thJA} value is according to JEDEC JESD51-3 at natural convection on FR4 1s0p board; the product (chip and package) was simulated on a 76.2 × 114.3 × 1.5 mm³ board with one copper layer (1 × 70 μm Cu).

4 Block description and electrical characteristics

4.1 Functional description tracking regulator

The regulator controls the output voltage V_{OUT} by comparing it to the voltage applied to the ADJ pin and driving a pLDMOS pass transistor accordingly. The stability of the control loop depends on:

- The output capacitor C_{OUT}
- Load current
- Chip temperature
- The poles and zeroes in the frequency response of the circuit consisting of the HE115(J) and the load

An input capacitor C_{IN} is strongly recommended for buffering the line influence.

To ensure stable operation, the output capacitor's capacitance and its equivalent series resistance ESR must fulfill the requirements in

Table 3-2 must be maintained. For details see also the typical performance graph Figure 4-12 . The output capacitor must be sized suitably to buffer load transients.

Connect each capacitor close to the pins.

The internal protection features are designed to protect the device itself as well as the application from destruction in case of catastrophic events. These safeguards contain:

- Output current limitation
- Reverse polarity protection
- Thermal shutdown

Output current limitation

In order to protect the pass element and the package from excessive power dissipation, the device limits the maximum output current at high input voltage.

Reverse polarity protection

The device allows a negative supply voltage.

Thermal shutdown

Block description and electrical characteristics

The overtemperature protection circuit is designed to prevent immediate destruction of the device in certain fault conditions (for example a permanent short circuit at output) by switching off the power stage. After the chip cools down, the regulator restarts. If the fault is not removed, then this leads to an oscillatory behavior of the output voltage. A junction temperature above 150°C is outside the maximum ratings and reduces the lifetime of the device.

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4.2 Electrical characteristics tracking regulator

$V_{IN} = 13.5 \text{ V}$; $2.0 \text{ V} \leq V_{ADJ} \leq 14 \text{ V}$; $V_{EN} \geq 2.0 \text{ V}$; $T_j = -40 \text{ }^{\circ}\text{C}$ to $150 \text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Table 4-1 Electrical characteristics tracking regulator

Symbol	Parameter	Condition	Value			Unit
			Min	Typ	Max	
ΔV_{OUT}	Output voltage tracking accuracy	$\Delta V_{OUT} = V_{ADJ} - V_{OUT}$; $5.5 \text{ V} \leq V_{IN} \leq 22 \text{ V}$; $0.1 \text{ mA} \leq I_{OUT} \leq 150 \text{ mA}$; $2 \text{ V} \leq V_{ADJ} \leq V_{IN} - 1 \text{ V}$	-5	-	5	mV
ΔV_{OUT}	Output voltage tracking accuracy	$\Delta V_{OUT} = V_{ADJ} - V_{OUT}$; $5.5 \text{ V} \leq V_{IN} \leq 32 \text{ V}$; $0.1 \text{ mA} \leq I_{OUT} \leq 70 \text{ mA}$; $2 \text{ V} \leq V_{ADJ} \leq V_{IN} - 1 \text{ V}$	-5	-	5	mV
$\Delta V_{OUT,load}$	Load regulation steady-state	$I_{OUT} = 0.1 \text{ mA}$ to 150 mA ; $V_{ADJ} = 5 \text{ V}$	-4	-0.1	-	mV
$\Delta V_{OUT,line}$	Line regulation steady-state	$V_{IN} = 5.5 \text{ V}$ to 32 V ; $I_{OUT} = 10 \text{ mA}$; $V_{ADJ} = 5 \text{ V}$	-	0.1	4	mV
PSRR	Power supply ripple rejection	$f_{ripple} = 100 \text{ Hz}$; ⁽¹⁾ $V_{ripple} = 1 \text{ Vpp}$; $I_{OUT} = 10 \text{ mA}$; $C_{OUT} = 10 \text{ } \mu\text{F}$, ceramic type	-	85	-	dB
$I_{OUT,max}$	Output current limitation	$V_{OUT} = V_{ADJ} - 0.1 \text{ V}$; $V_{ADJ} = 5 \text{ V}$	151	350	500	mA
$I_{OUT,rev}$	Reverse current	$V_{IN} = 0 \text{ V}$; $V_{OUT} = 16 \text{ V}$; $V_{ADJ} = 5 \text{ V}$	-50	-16	-	μA
$I_{IN,rev}$	Reverse current at negative input voltage	$V_{IN} = -16 \text{ V}$; $V_{OUT} = 0 \text{ V}$; $V_{ADJ} = 5 \text{ V}$	-10	0	-	μA

HE115(J)

High-precision voltage tracker



Block description and electrical characteristics

V_{dr}	Dropout voltage	$V_{dr} = V_{IN} - V_{OUT};^{(2)}$ $I_{OUT} = 150 \text{ mA};$ $V_{ADJ} = 5 \text{ V}$	-	250	500	mV
$T_{j,sd}$	Overtemperature shutdown threshold	T_j increasing due to power dissipation generated by the device	-	175	-	°C
$\Delta T_{j,sdh}$	Overtemperature shutdown threshold hysteresis		-	15	-	K

(1) Not subject to production test, specified by design.

(2) Measured when the output voltage V_{OUT} has dropped 100 mV from the nominal value obtained at $V_{IN} = 13.5 \text{ V}$.

4.3 Typical performance characteristics tracking regulator.

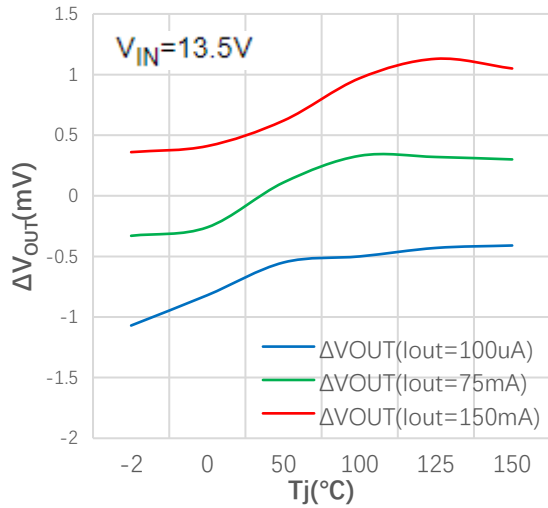


Figure 4-1 Tracking accuracy ΔV_{OUT} versus junction temperature T_j

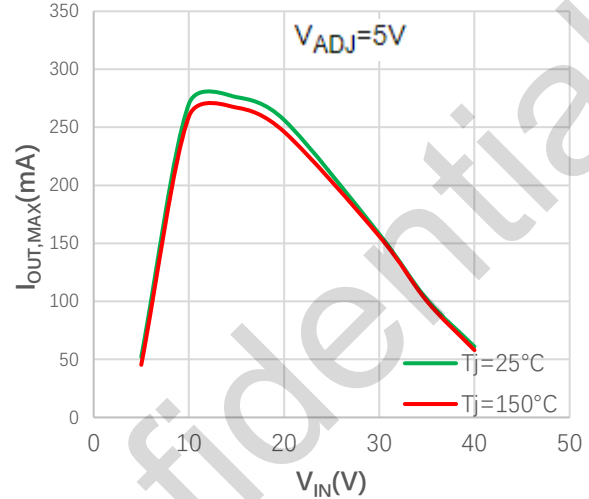


Figure 4-2 Output current limitation $I_{OUT,MAX}$ versus input voltage V_{IN}

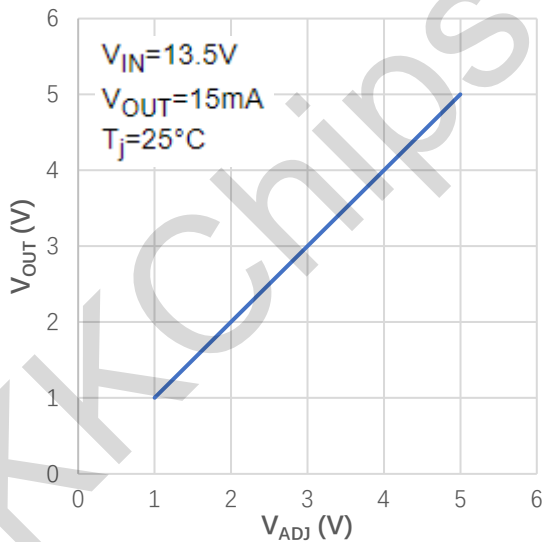


Figure 4-3 Output voltage V_{OUT} versus adjust voltage V_{ADJ}

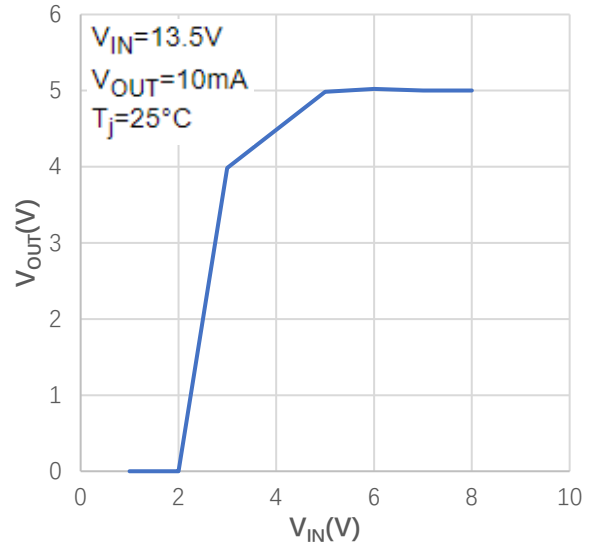


Figure 4-4 Output voltage V_{OUT} versus input voltage V_{IN}

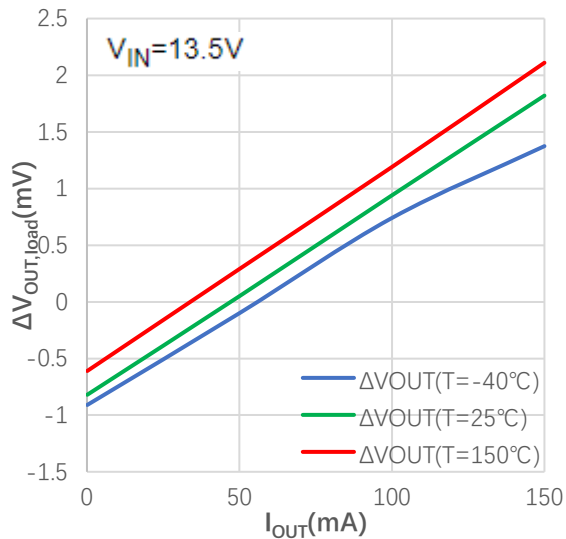


Figure 4-5 Load regulation $\Delta V_{OUT,load}$ versus output current I_{OUT}

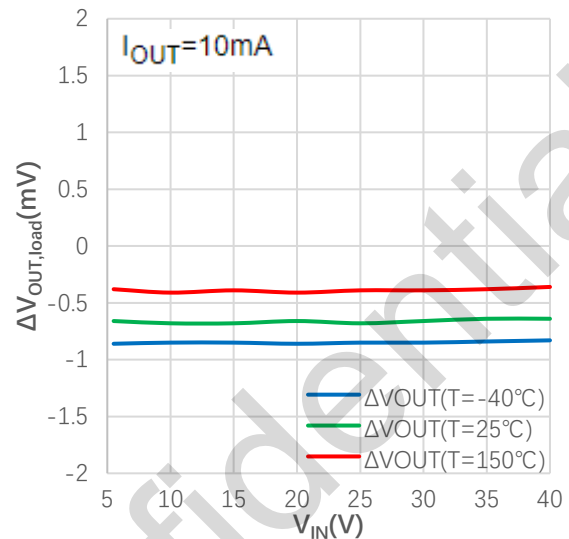


Figure 4-6 Line regulation $\Delta V_{OUT,line}$ versus input voltage V_{IN}

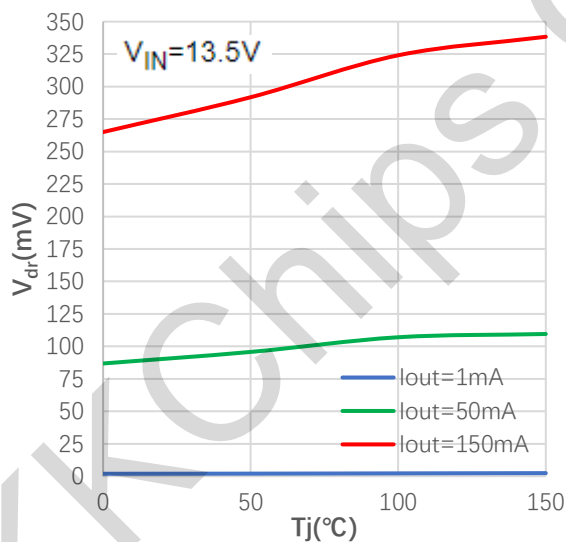


Figure 4-7 Dropout voltage V_{dr} versus junction temperature T_j

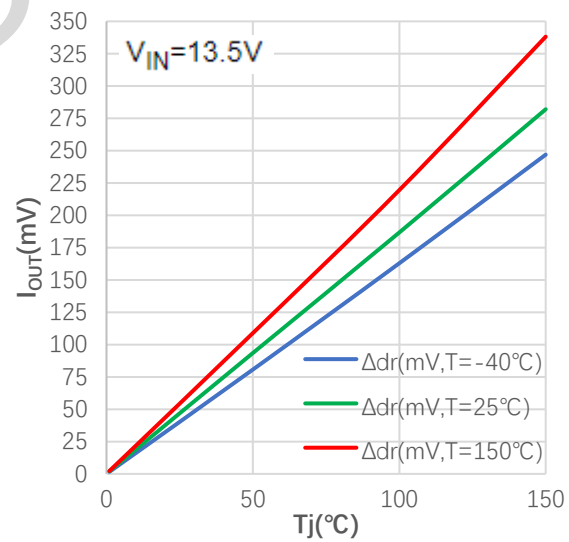


Figure 4-8 Dropout voltage V_{dr} versus output current I_{OUT}

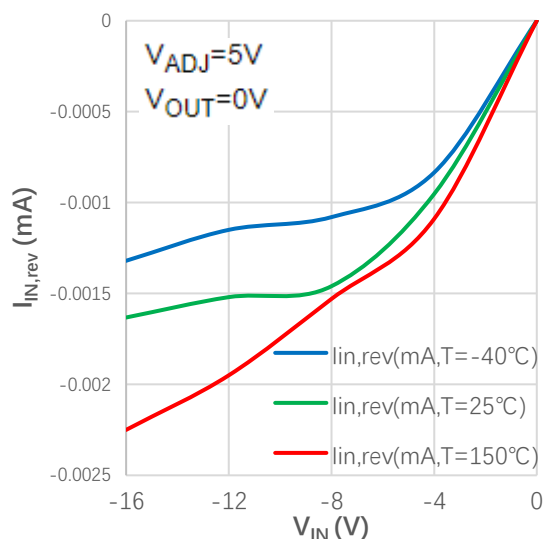


Figure 4-9 Reverse current $I_{IN,rev}$ versus input voltage V_{IN}

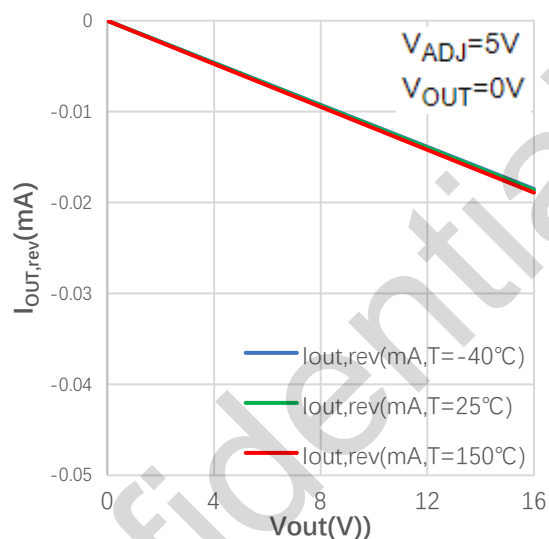


Figure 4-10 Reverse current $I_{OUT,rev}$ versus output voltage V_{OUT}

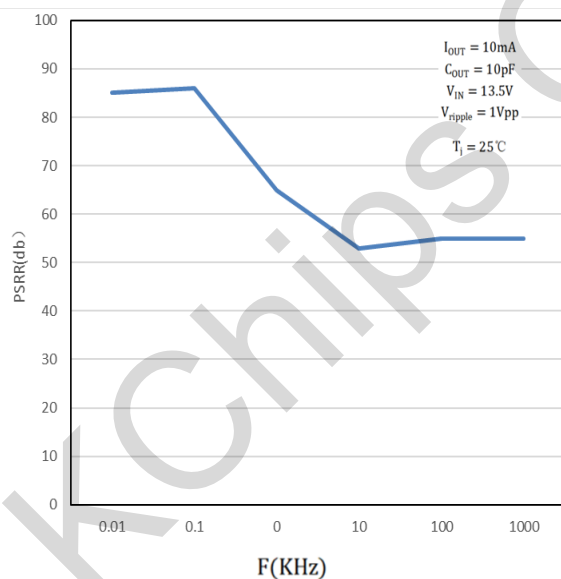


Figure 4-11 Power supply ripple rejection PSRR versus ripple frequency f_r

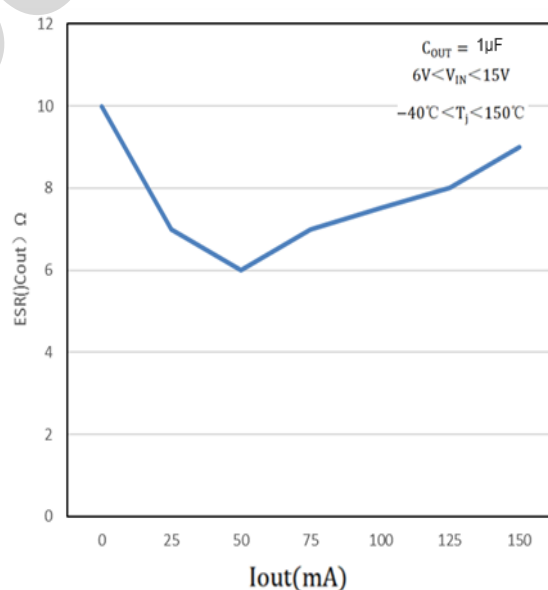


Figure 4-12 Output capacitor ESR(C_{OUT}) versus output current I_{OUT}

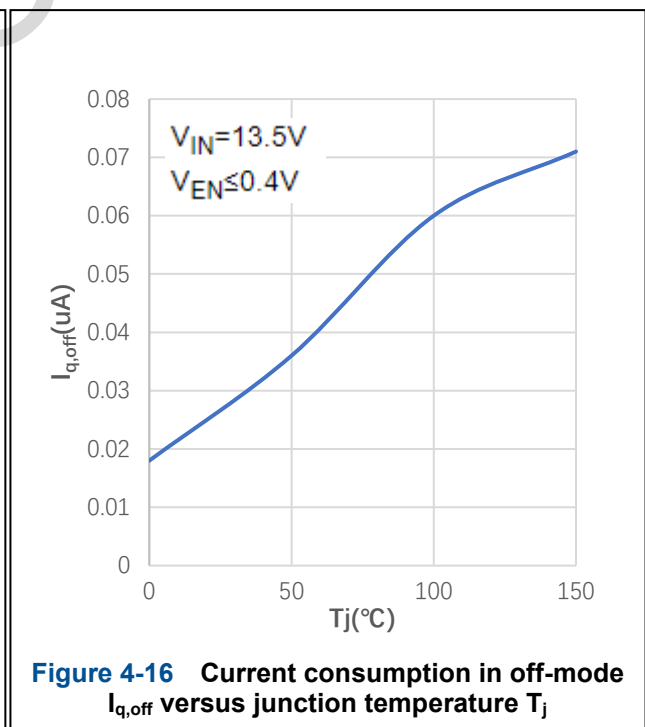
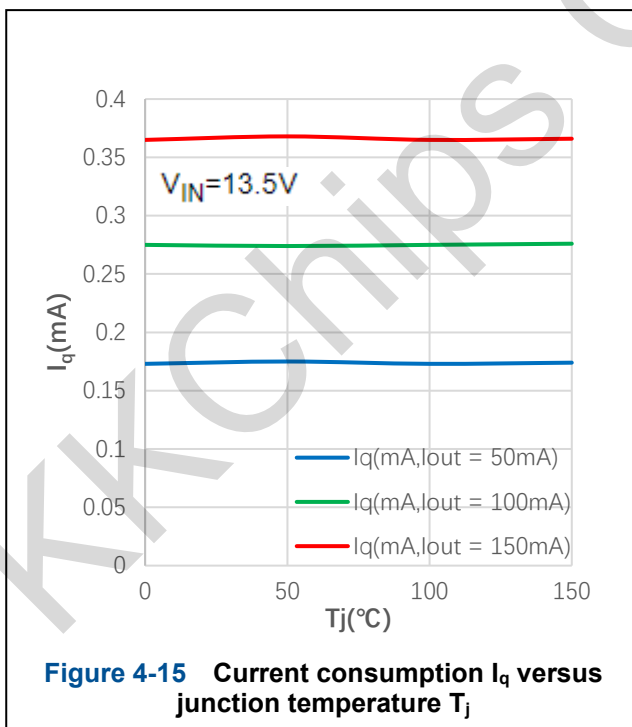
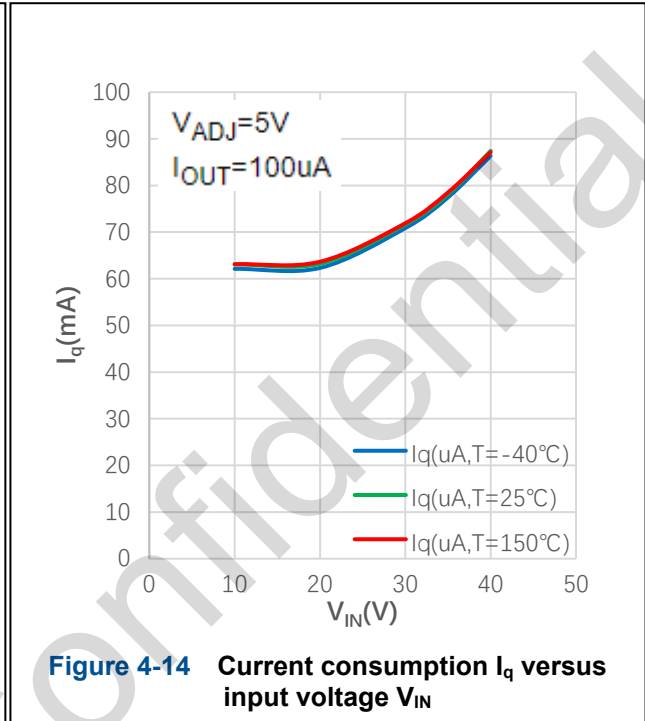
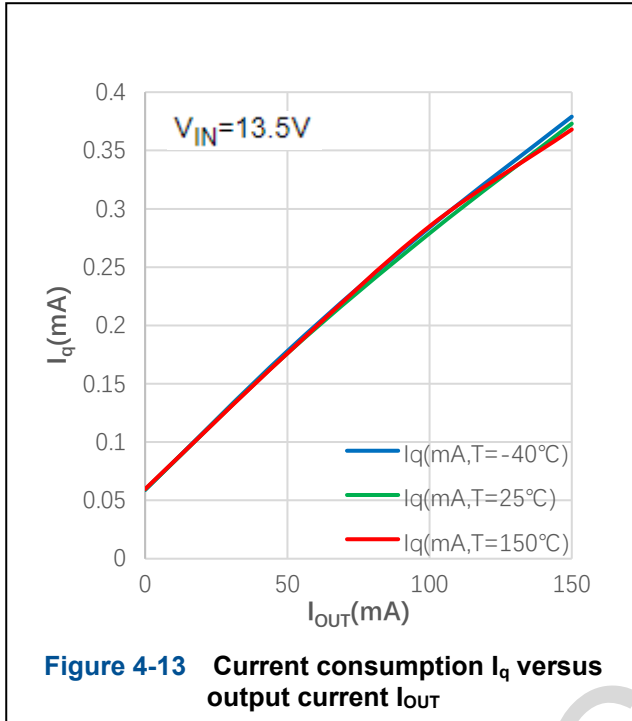
4.4 Electrical characteristics current consumption

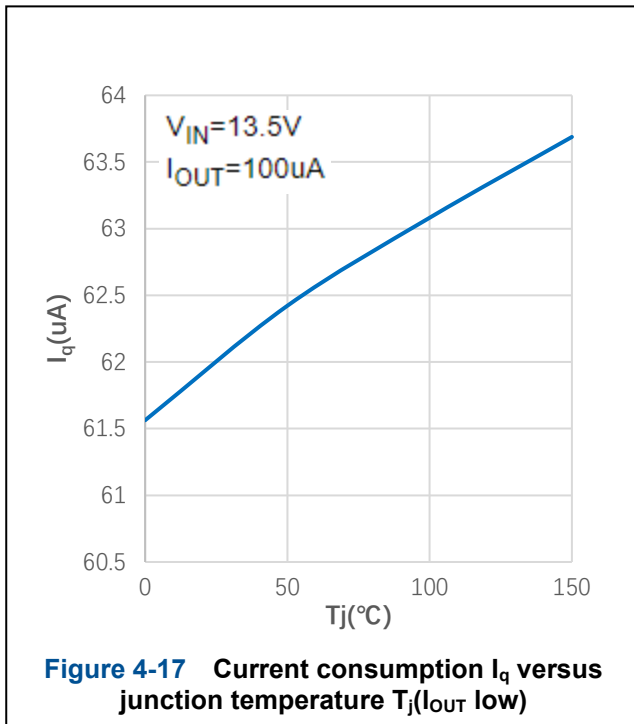
Table 4-2 Electrical characteristics current consumption

Symbol	Parameter	Condition	Value			Unit
			Min	Typ	Max	
$I_{q,off}$	Current consumption standby mode	$I_{q,off} = I_{IN};$ $V_{EN} \leq 0.4 \text{ V};$ $T_j \leq 125 \text{ }^{\circ}\text{C}$	-	0.1	5	μA
I_q	Current consumption	$I_q = I_{IN} - I_{OUT};$ $I_{OUT} \leq 0.1 \text{ mA};$ $V_{ADJ} = 5 \text{ V};$ $T_j \leq 125 \text{ }^{\circ}\text{C}$	-	55	90	μA
I_q	Current consumption	$I_q = I_{IN} - I_{OUT};$ $I_{OUT} \leq 150 \text{ mA};$ $V_{ADJ} = 5 \text{ V}$	-	0.3	0.5	mA

$V_{IN} = 13.5 \text{ V}; 2.0 \text{ V} \leq V_{ADJ} \leq 14 \text{ V}; V_{EN} \geq 2.0 \text{ V}; T_j = -40 \text{ }^{\circ}\text{C}$ to $150 \text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

4.5 Typical performance characteristics current consumption





4.6 Functional description enable input

On a "low" signal at the enable input EN the device switches to standby mode in order to minimize the quiescent current.

If the EN pin is not connected, then the "low" level from the internal pull-down resistor switches off the regulator.

4.7 Electrical characteristics enable input

$V_{IN} = 13.5 \text{ V}$; $2.0 \text{ V} \leq V_{ADJ} \leq 14 \text{ V}$; $V_{EN} \geq 2.0 \text{ V}$; $T_j = -40 \text{ }^{\circ}\text{C}$ to $150 \text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Table 4-3 Electrical characteristics enable input

Symbol	Parameter	Condition	Value			Unit
			Min	Typ	Max	
$V_{EN,off}$	Enable off voltage range	$V_{OUT} = 0 \text{ V}$; $I_{OUT} \leq 5 \text{ } \mu\text{A}$; $T_j \leq 125^{\circ}\text{C}$	-	-	0.8	V
$V_{EN,on}$	Enable on voltage range	V_{OUT} settled	2	-	-	V

Block description and electrical characteristics

I_{EN}	Enable input current	$V_{EN} = 5\text{ V}$	-	2	4	μA
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4.8 Typical performance characteristics enable input

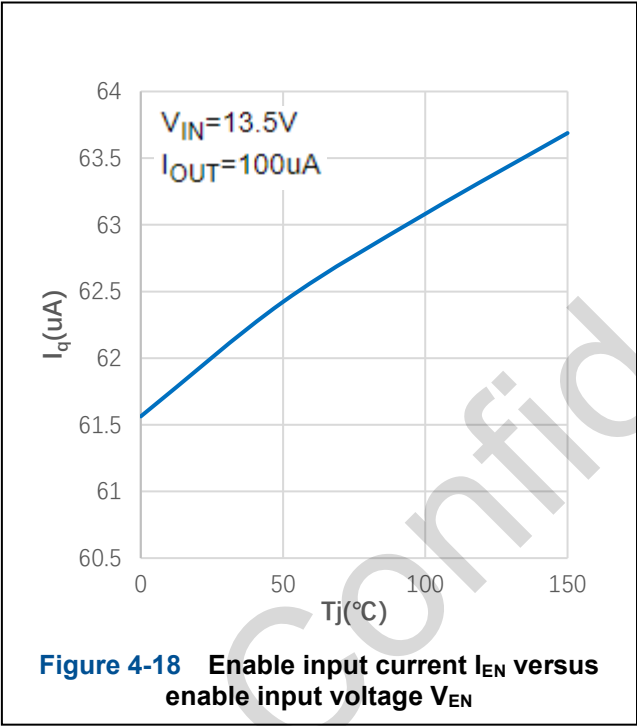


Figure 4-18 Enable input current I_{EN} versus enable input voltage V_{EN}

4.9 Functional description adjust input

The adjust input must be connected to the reference voltage that the device tracks.

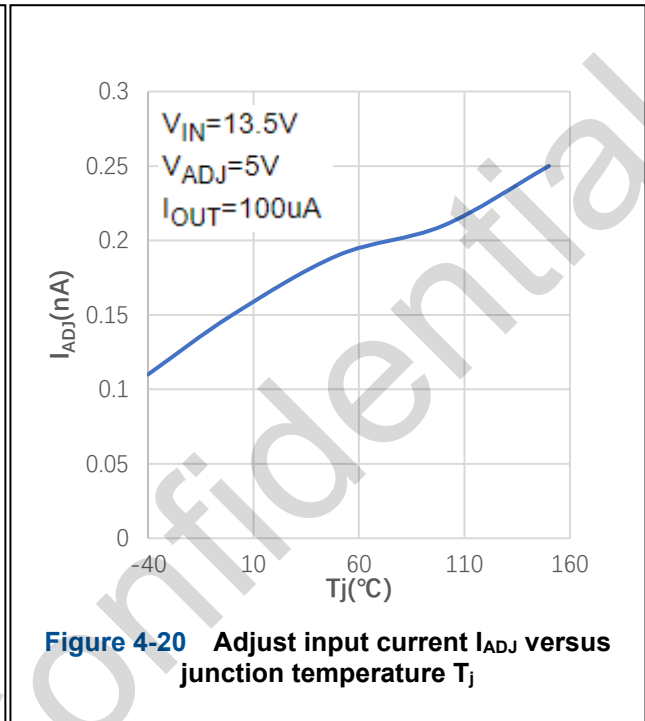
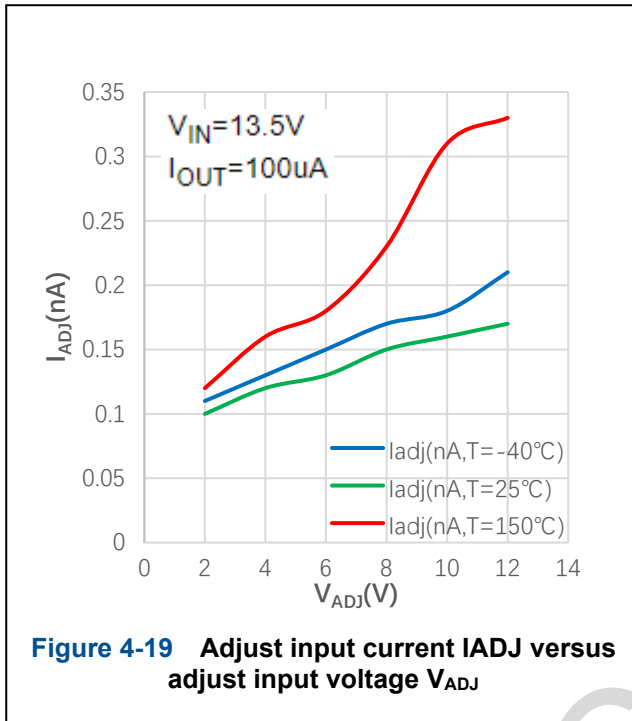
4.10 Electrical characteristics adjust input

$V_{IN} = 13.5\text{ V}$; $2.0\text{ V} \leq V_{ADJ} \leq 14\text{ V}$; $V_{EN} \geq 2.0\text{ V}$; $T_j = -40\text{ }^\circ\text{C}$ to $150\text{ }^\circ\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Table 4-4 Electrical characteristics adjust input

Symbol	Parameter	Condition	Value			Unit
			Min	Typ	Max	
I_{ADJ}	Adjust input current	$V_{ADJ} = 5\text{ V}$	-	0.03	1	μA

4.11 Typical performance characteristics adjust input.



4.12 Power Good Output

The Power Good output PG indicates an overvoltage or undervoltage condition of the tracker output. For this the HE115(J) compares the output voltage V_{OUT} to the reference voltage V_{ADJ} . Variations of the output voltage beyond the Power Good switching thresholds are indicated by a “low” signal at the Power Good output PG.

Transients shorter than the Power Good reaction time $t_{PG,r}$ do not trigger the Power Good output.

The Power Good output PG is an open collector output that requires a pull-up resistor to a positive voltage rail. The pull-up voltage must not exceed the absolute maximum ratings of Power Good PG (see “Absolute maximum ratings” on Page 9).

4.13 Electrical characteristics Power Good Output

$V_{IN} = 13.5\text{ V}$; $2.0\text{ V} \leq V_{ADJ} \leq 14\text{ V}$; $V_{EN} \geq 2.0\text{ V}$; $T_j = -40\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Table 4-5 Electrical characteristics Power Good output

Symbol	Parameter	Condition	Value			Unit
			Min	Typ	Max	
$V_{OUT,UV}$	Power Good switching threshold, undervoltage	V_{OUT} decreasing, $V_{IN} \geq V_{ADJ} + 150\text{mV}$	$V_{ADJ} - 120$	$V_{ADJ} - 70$	$V_{ADJ} - 50$	mV
$V_{OUT,OV}$	Power Good switching threshold, overvoltage	V_{OUT} increasing, $V_{IN} \geq V_{ADJ} + 150\text{mV}$	$V_{ADJ} + 50$	$V_{ADJ} + 70$	$V_{ADJ} + 120$	mV
$t_{PG,r}$	Power Good reaction time	-	10	15	30	μs
$V_{PG,LOW}$	Power Good output low voltage	$V_{IN} \geq 4\text{V}$, $I_{PG,ext} \leq 1.8\text{mA}$	-	0.2	0.4	V
$I_{PG,ext}$	Power Good output external input current	$V_{PG} \leq 0.4\text{V}$	-	-	1.8	mA
$I_{PG,leak}$	Power Good output leakage current	$V_{OUT} = V_{ADJ}$, $V_{PG} = 5\text{V}$	-	0	2	μA

5 Application information

The following information is given as an example for the implementation of the device only and shall not be regarded as a description or warranty of a certain functionality, condition or quality of the device.

5.1 Application diagram

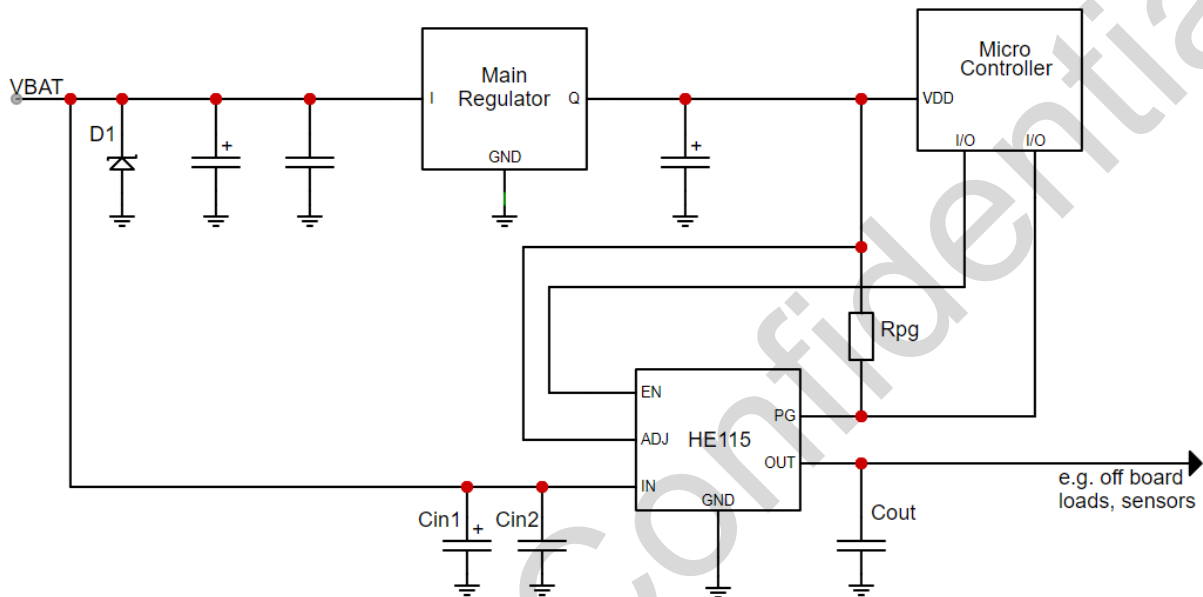


Figure 5-1 Application diagram

This figure is a simplified example of an application circuit. The function must be verified in the application.

5.2 Selection of external components

5.2.1 Input pin

Figure 5-1 shows the typical input circuitry for a voltage tracking regulator. The following external components at the input are recommended in case of possible external disturbance.

Ceramic capacitor

A ceramic capacitor C_{IN1} (100 nF to 470 nF) at the input filters high frequency disturbance imposed by the line, such as ISO pulses 3a/b. Place C_{IN1} as close as possible to the input pin of the voltage tracking regulator on the PCB.

Aluminum electrolytic capacitor

An aluminum electrolytic capacitor C_{IN2} (10 μ F to 470 μ F) at the input smoothens high energy pulses, such as ISO pulse 2a. Place C_{IN2} close to the input pin of the voltage tracking regulator on the PCB.

Overvoltage suppression diode

A suitably sized diode D1 suppresses high voltage beyond the maximum ratings of the circuit components and protects the device from damage due to overvoltage.

5.2.2 Output pin

The requirements for C_{OUT} are described in the Table 3-2. The Figure 4-12 shows the stable operation range of the device.

In an automotive environment, ceramic capacitors with X5R or X7R dielectrics are recommended.

Place C_{OUT} on the same side of the PCB as the device and as close as possible to both the OUT pin and the GND pin.

In case of rapid transients of input voltage or load current, C_{OUT} must be dimensioned accordingly to ensure the output stability in the application.

5.2.3 Adjust pin

Figure 5-1 shows a typical adjust circuitry for a voltage tracking regulator. Typically the adjust pin is

connected to a fixed voltage reference that the regulator tracks. In the example of the application diagram ADJ is connected to the supply voltage of a microcontroller. Alternatively, the voltage reference can also be adjusted by a voltage divider.

5.2.4 Power Good pin

The Power Good output is an open collector output, which requires a pull-up resistor to a positive voltage rail. The pull-up voltage must not exceed the maximum ratings of the Power Good PG shown in “Absolute maximum ratings” on [Page 9](#). In [Figure 5-1](#) the supply voltage VDD of a microcontroller is used as pull-up voltage for example.

To limit the external input current according to the requirement (see “Power Good Output” on [Page 23](#)), the resistor must be sized depending on the pull-up voltage.

5.3 Thermal considerations

From the known input voltage, the output voltage and the load profile of the application, the total power dissipation can be calculated:

$$P_D = (V_{IN} - V_{OUT}) \times I_{OUT} + V_{IN} \times I_q$$

Equation 1

with

- P_D : continuous power dissipation
- V_{IN} : input voltage
- V_{OUT} : output voltage
- I_{OUT} : output current
- I_q : quiescent current

The maximum acceptable thermal resistance R_{thJA} can then be calculated:

$$R_{thJA,max} = \frac{T_{j,max} - T_a}{P_D}$$

Equation 2

with

- $T_{j,max}$: maximum allowed junction temperature
- T_a : ambient temperature

Based on the above calculation the proper PCB type and the necessary heat sink area can be determined with reference to the specification in [Table 3-3](#).

Example

Application conditions:

$$V_{IN} = 13.5 \text{ V}$$

$$V_{OUT} = V_{ADJ} = 5 \text{ V}$$

$$I_{OUT} = 100 \text{ mA}$$

$$T_a = 75^\circ\text{C}$$

Calculation of $R_{thJA,max}$:

$$\begin{aligned} P_D &= (V_{IN} - V_{OUT}) \times I_{OUT} + V_{IN} \times I_q \\ &= (13.5\text{V} - 5\text{V}) \times 100 \text{ mA} + 13.5 \text{ V} \times 3.5 \text{ mA} \\ &= 0.897 \text{ W} \end{aligned}$$

$$\begin{aligned} R_{thJA,max} &= (T_{j,max} - T_a) / P_D \\ &= (150^\circ\text{C} - 75^\circ\text{C}) / 0.897 \text{ W} \\ &= 83.6 \text{ K/W} \end{aligned}$$

As a result, the PCB design must ensure a thermal resistance R_{thJA} lower than 83.6 K/W. According to "Thermal resistance" on [Page 10](#), at least 300 mm² heatsink area is required on the FR4 1s0p PCB, or the FR4 2s2p board can be used.

6 Package and packing information.

6.1 ESOP-8 mechanical data

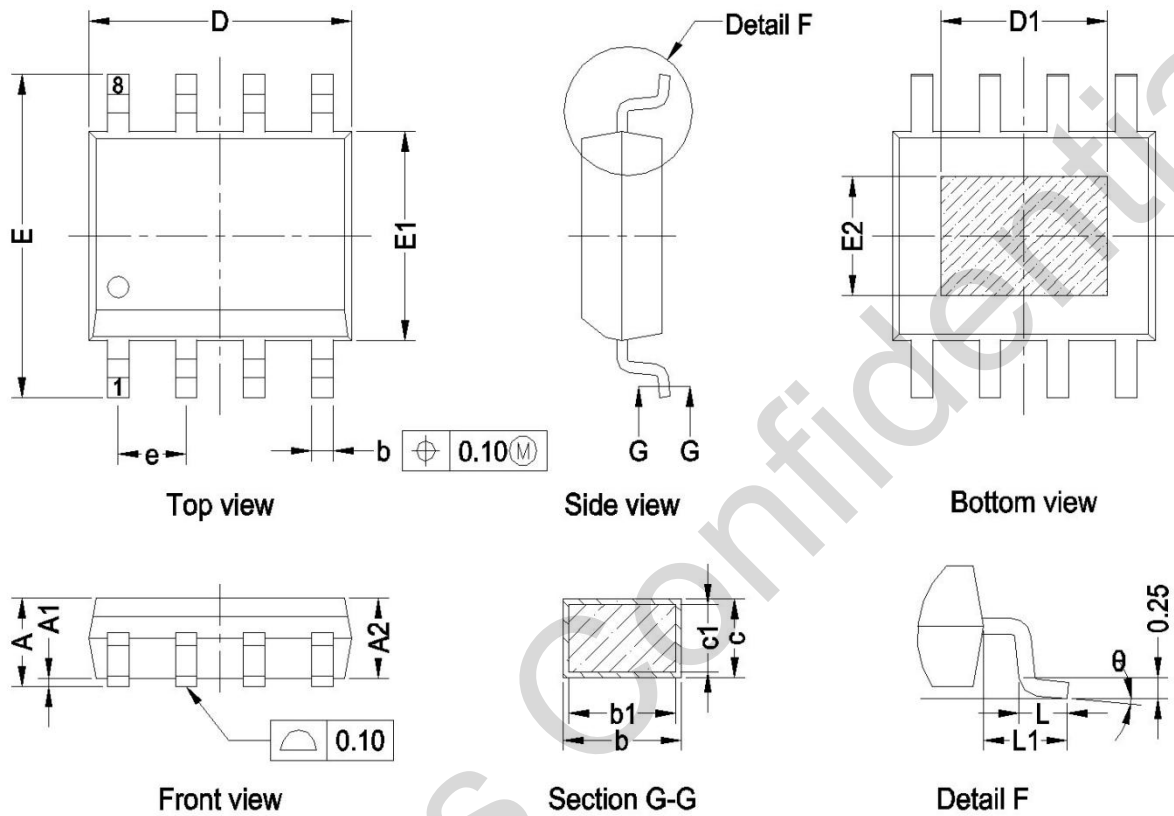


Figure 6-1 ESOP-8 package dimension

Table 6-1 ESOP-8 mechanical data

Dim.	mm		
	Min.	Nom	Max.
A	-	-	1.70
A1	0.05	0.10	0.15
A2	1.30	1.45	1.55
b	0.39	-	0.51
b1	0.36	0.41	0.46
c	0.17	-	0.25
c1	0.203 REF		
D	4.70	4.90	5.10
D1	3.00	-	3.25
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
E2	2.10	-	2.35
e	1.27 BSC		
L	0.40	0.6	0.80
L1	1.00 REF		
θ	0°	-	8°

6.2 DFN-10L mechanical data

Package and packing information

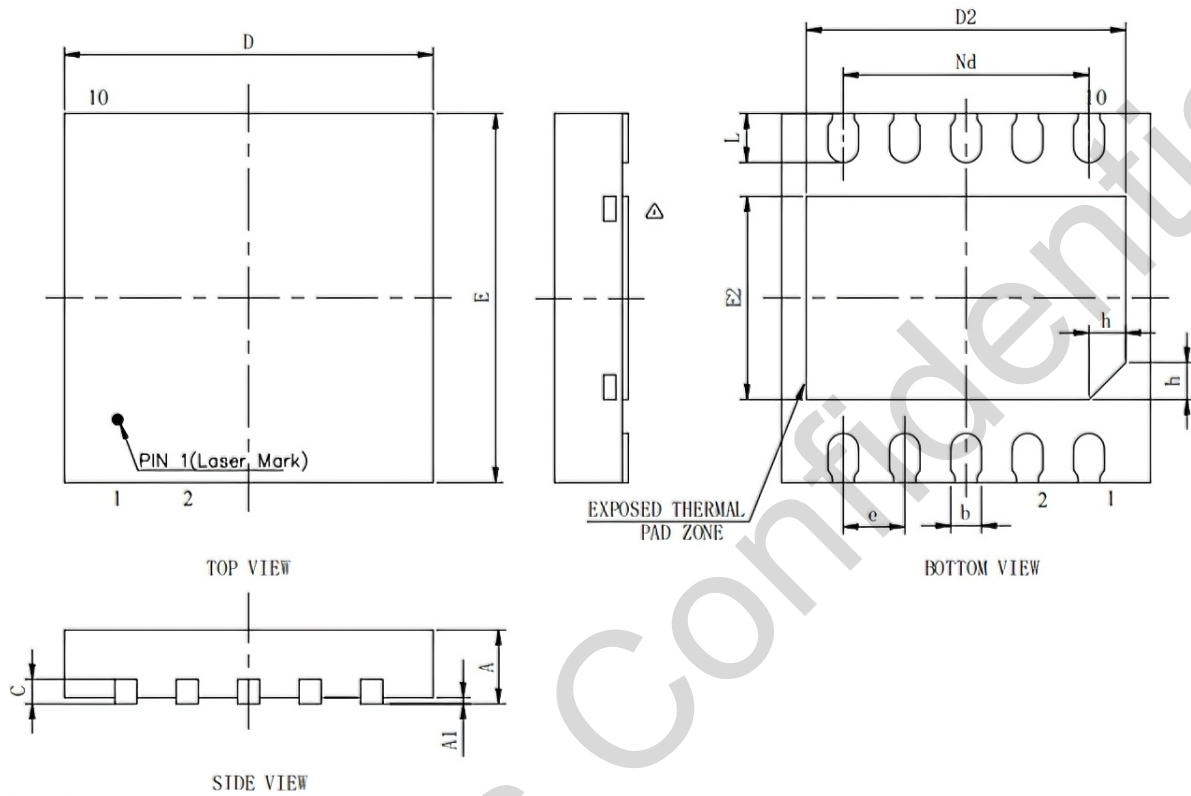


Figure 6-2 DFN-10L package dimension

Table 6-2 DFN-10L mechanical data

Dim.	mm		
	Min.	Nom	Max.
A	0.85	0.90	0.95
A1	0	0.02	0.05
b	0.20	0.25	0.30
b1	0.18REF		
c	0.203REF		
D	2.90	3.00	3.10
D2	2.40	2.50	2.60
e	0.50BSC		
Nd	2.00BSC		
E	2.90	3.00	3.10
E2	1.45	1.55	1.65
L	0.30	0.40	0.50
h	0.20	0.25	0.30

7 Recommended pad size

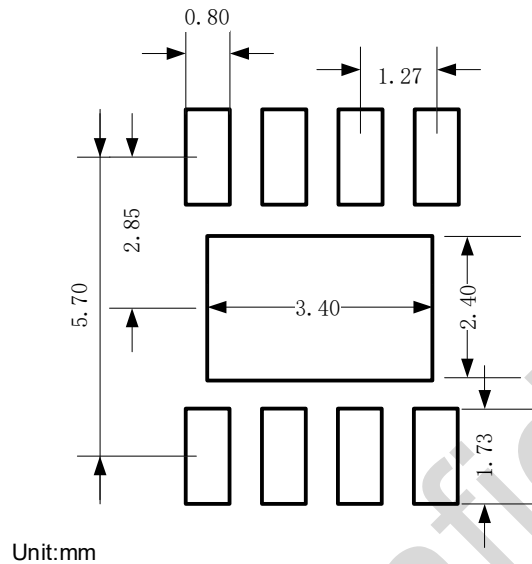


Figure 7-1 ESOP-8 Recommended pad size

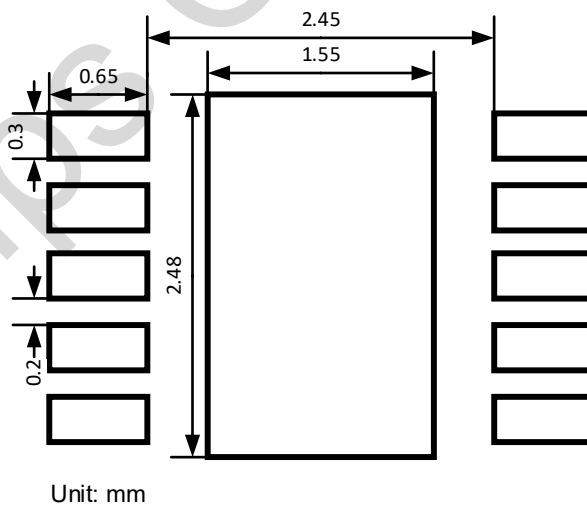


Figure 7-2 DFN-10L Recommended pad size

8 Customer Service

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9 Revision history

Date	Revision	Changes
2022/10/11	1.1	1.Updated "Note *" in the Section 2 Electrical specifications; 2.Updated Figure 2-1 Current and voltage conventions.
2023/03/13	1.2	1.Update typography; 2.Updated Part number; 3.Update charts; 4.Remove SO-8 related data.
2023/09/12	1.3	1.Format updated to 1.3
2024/02/05	1.4	1.Format updated to 1.4.
2024/04/22	1.6	1.Package information update. 2. Format updated to 1.6.
2024/04/22	1.5	1.Package information update. 2.Format updated to 1.5.
2024/05/14	1.7	1.Modify the captions in Table 4-3, 4-4, 4-5. 2.Modify the captions in Figure 4-17
2024/05/20	1.8	1. Change the package diagram and package information sequence
2024/05/21	1.9	1.Add pad size.
2024/08/01	2.0	1.Modify Figure 4-12.
2024/09/03	2.1	1. Updated the company's new address
2024/12/09	2.2	1.Format updated to 2.2 2.Modify the error description of Vout and ov in Table 4-5.
2025/2/7	2.3	1.Optimize Figure 1-1.
2025/4/11	2.4	1. Revise the captions in Figure 4-7.
2025/10/31	2.5	1. Add DFN-10L package information.